Lab: Multiple Linear Regression Analysis

Estimated time needed: 30 minutes

In this lab, you'll learn how to perform a multiple linear regression analysis using Microsoft Excel.

Software used in this lab

The instruction videos in this course will be using the full Excel Desktop version because it has all the available product features. However, for the hands-on labs, we will use the free "Excel for the web" version because this is available to everyone.

Although you can use the Excel Desktop software, it is recommended that you use Excel for the web for the hands-on labs, as the lab instructions specifically refer to this version. There are some small differences in the interface and available features.

Data set Used in this Lab

The data set used in this lab is curated from the following source: https://www.kaggle.com/datasets/yasserh/housing-prices-dataset under the CC0: Public Domain license.

Acknowledgment and thanks also go to https://trak.in, who were generous enough to share the data publicly for free.

Note: We are using a modified subset of the original data set. Please ensure you use the data set provided in this lab.

Learning objectives:

- Perform a multiple linear regression analysis.
- Assess how well the regression equation predicts the dependent variable.
- Assess the contribution of each independent variable to the prediction.

Welcome to the Lab: Multiple Linear Regression Analysis!

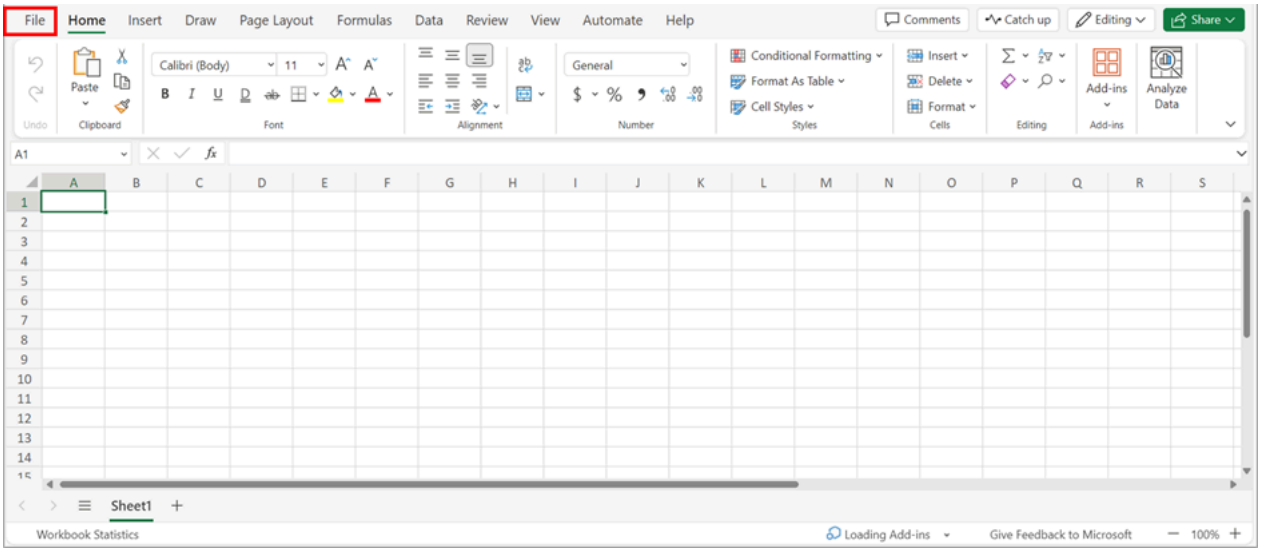
Configure the XL Miner Analysis ToolPak

Before you begin with the lab, let's configure the XL Miner Analysis ToolPak in your Excel for the web version. You will need this add-in to perform the regression analysis. If you have already configured the add-in, you may skip this section.

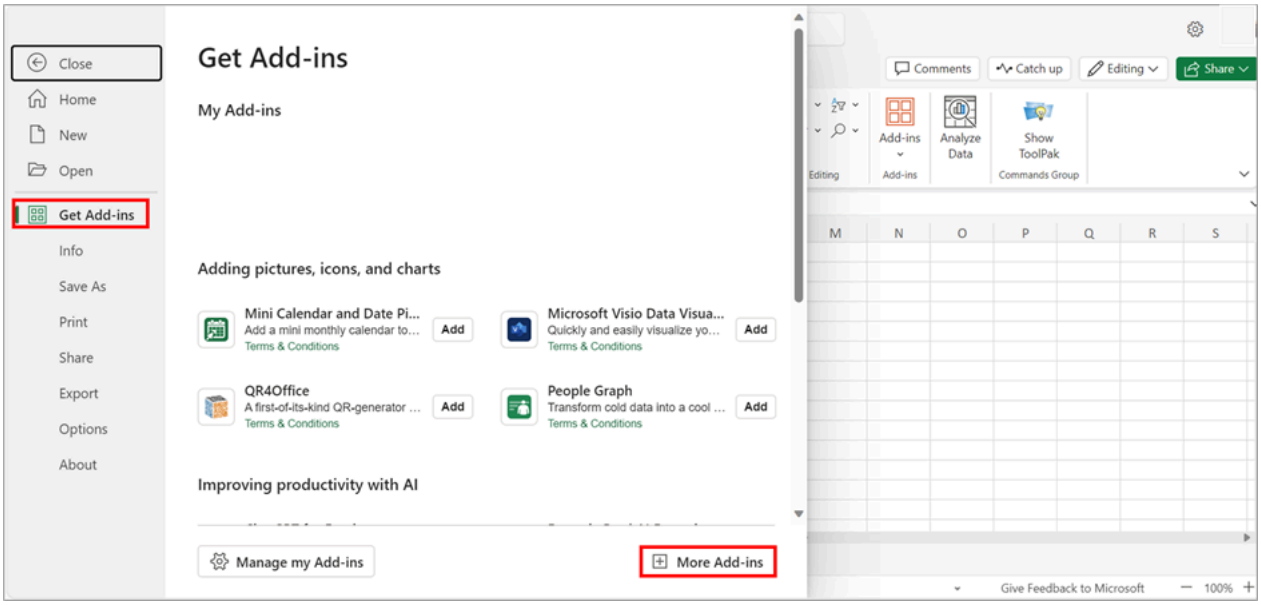
Note: XLMiner Analysis ToolPak is a third-party free statistical analysis ToolPak in Excel for the web. It shows the same functions as the Analysis ToolPak add-in available in Excel for the desktop. You can use this ToolPak to perform descriptive statistics and linear regression for the labs in this course. If you are using Excel for the desktop, you can configure the in-built Data Analysis ToolPak available from Microsoft.

To configure this add-in, open a file in Excel for the web and follow the following steps.

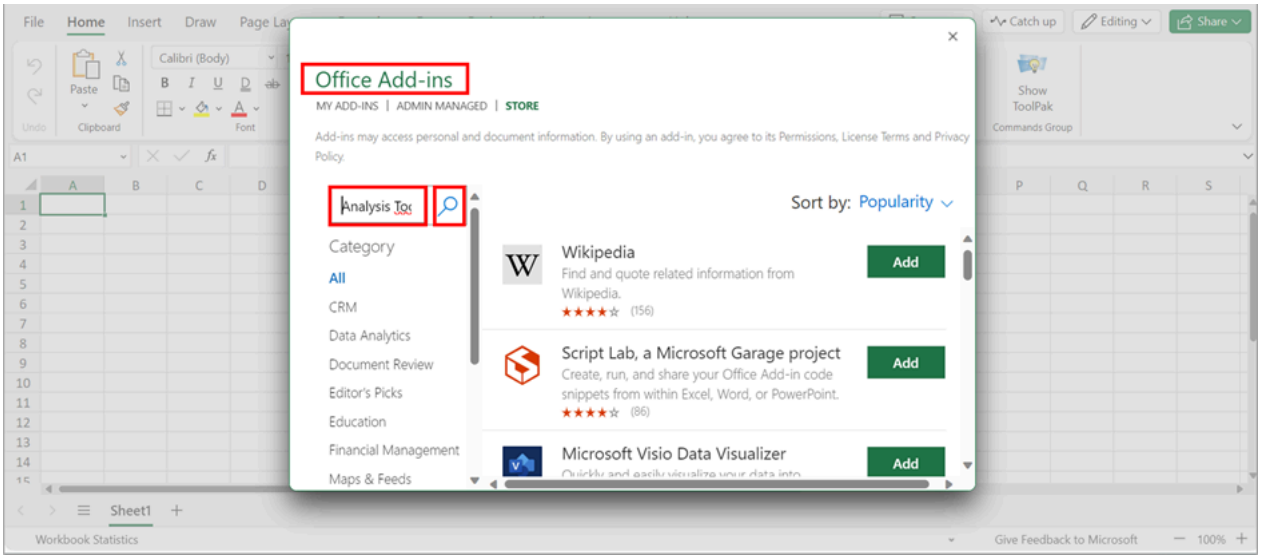
1. Select **File** on the top ribbon.



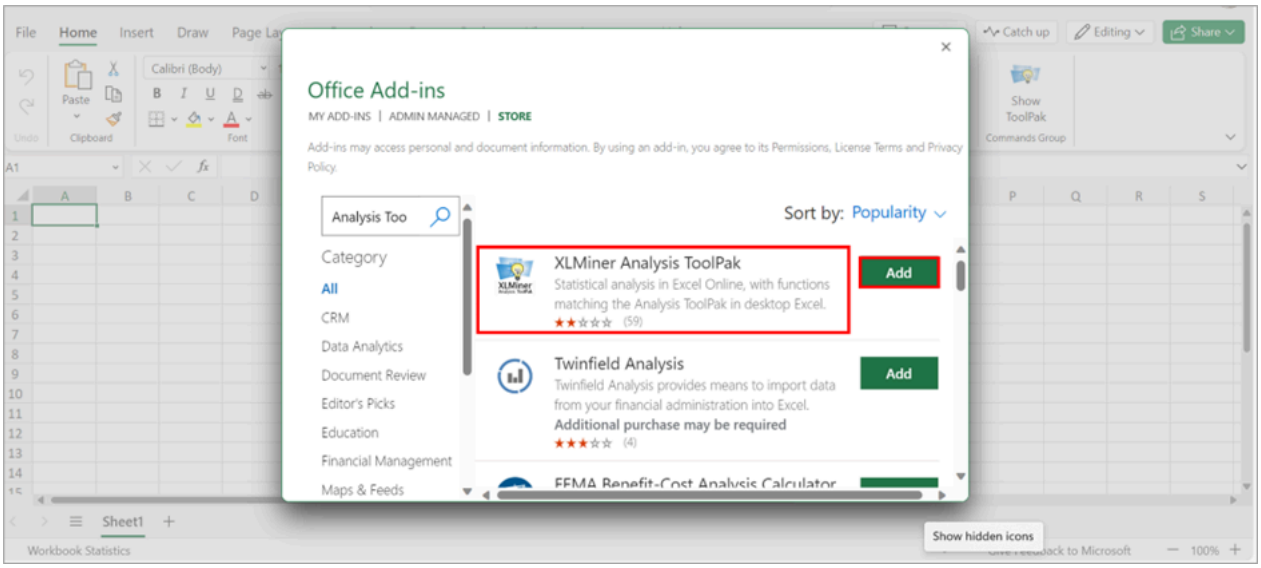
2. Select Get **Add-ins** from the left menu bar and choose **More Add-ins**.



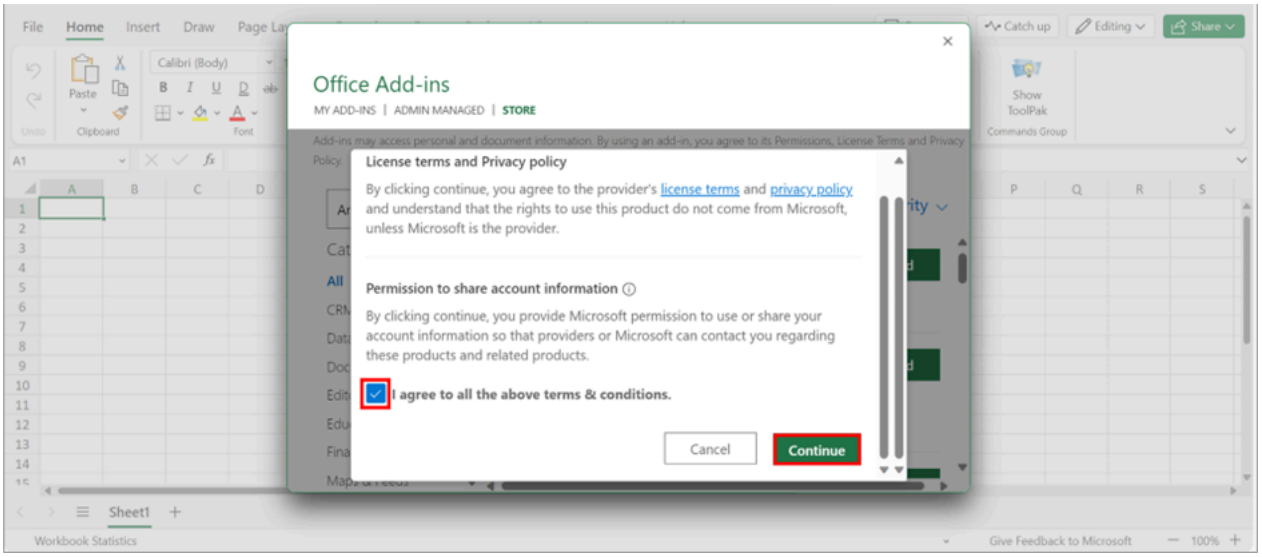
3. In the **Office Add-ins** window, type in **Analysis ToolPak** in the search bar and select the search icon.



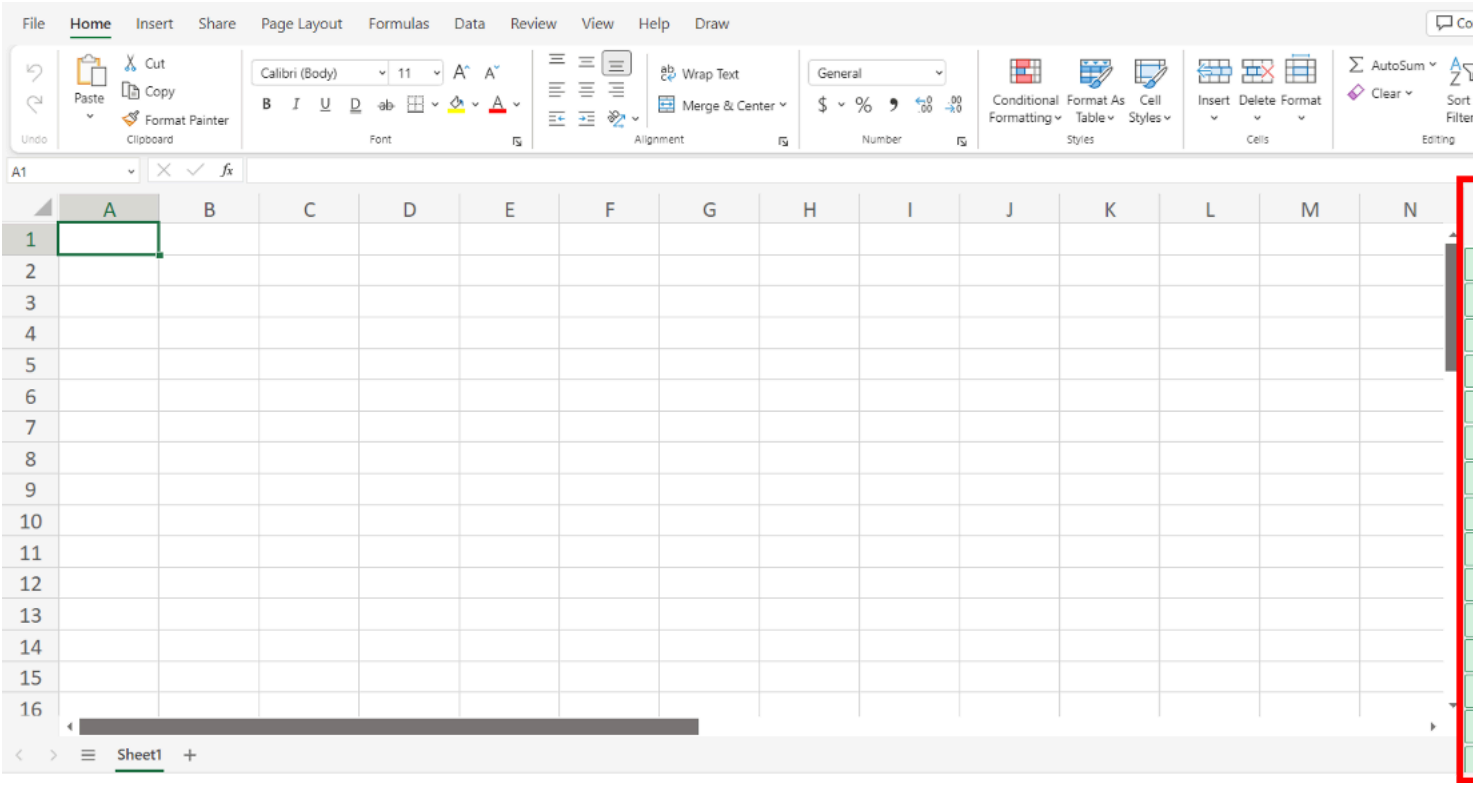
4. Select the **Add** button next to the **XLMiner Analysis ToolPak**.



5. Select the checkbox next to **I agree to all the above terms & conditions**. Select **Continue**.



6. You'll see **XLMiner Analysis ToolPak** on the right side of your Excel sheet. You'll also see the **Show ToolPak** button in the ribbon.

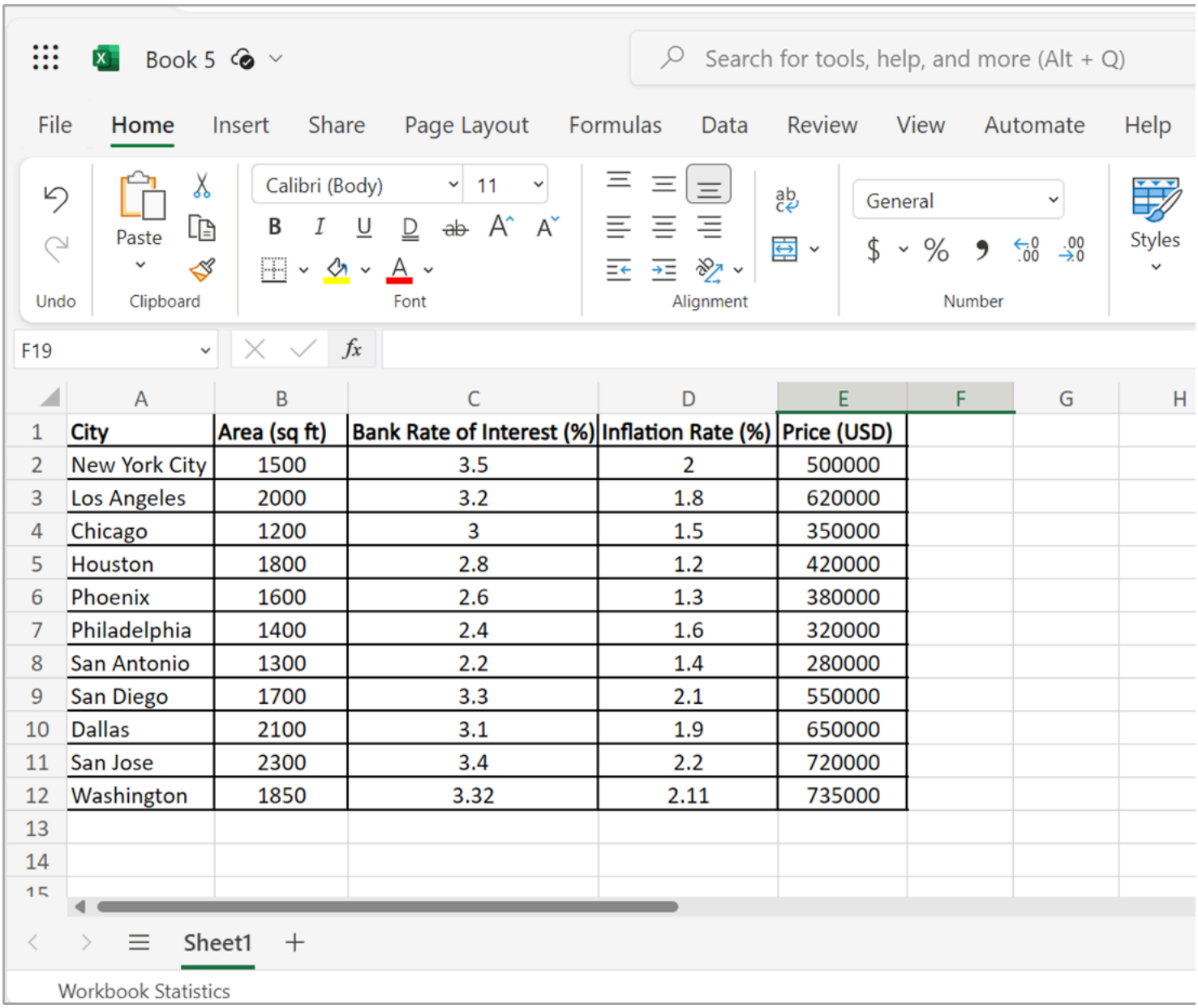


Exercise: Perform a multiple linear regression analysis to predict the test score

In this exercise, you'll learn to identify dependent and independent variables, perform multiple linear regression, evaluate the regression model, and assess the contribution of each independent variable.

For this lab, you will use the following data set, Lab_Multiple Linear Regression_Data set.xlsx. To download the data set, right-click the hyperlink and select Open link in a new tab. Open the file in Excel for the Web version and follow the steps given below to perform the regression analysis.

Here's a screenshot of the data set.



City	Area (sq ft)	Bank Rate of Interest (%)	Inflation Rate (%)	Price (USD)
New York City	1500	3.5	2	500000
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Dallas	2100	3.1	1.9	650000
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Washington	1850	3.32	2.11	735000

Task 1: Identify the dependent and independent variables

In this task, you'll predict the house prices, which depend on three variables: the area of the house in square feet, the inflation rate, and the bank's rate of interest. Thus, house prices are the dependent variable, whereas the area of the house in square feet, the inflation rate, and the bank's rate of interest are the independent variables.

You will use the following multiple regression equation with three independent variables:

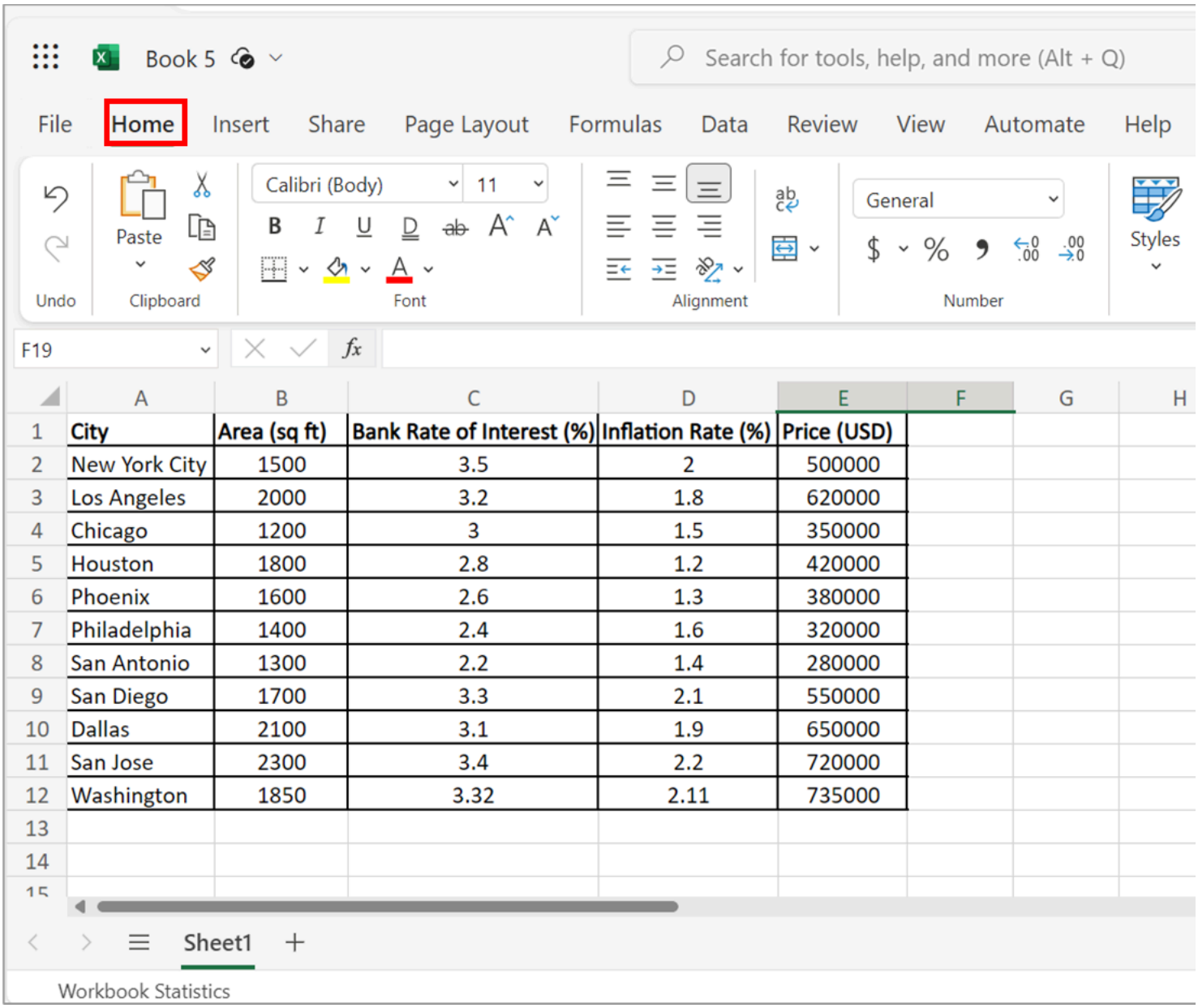
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$$

Where
Y: House price (Dependent variable)
X_1: Area of the house in square feet (Independent variable)
X_2: The inflation rate (Independent variable)
X_3: The interest rate (Independent variable)
β_0: The y-intercept
β_1, β_2, and β_3: The slope coefficients

Task 2: Perform the multiple linear regression

In this task, you'll perform regression analysis using Excel.

1. You'll be using the XL Miner Analysis Toolpak to perform this task. To open the add-in, Click the **Show ToolPak** icon within the **Home** tab.



	A	B	C	D	E	F	G	H
1	City	Area (sq ft)	Bank Rate of Interest (%)	Inflation Rate (%)	Price (USD)			
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11	San Jose	2300	3.4	2.2	720000			
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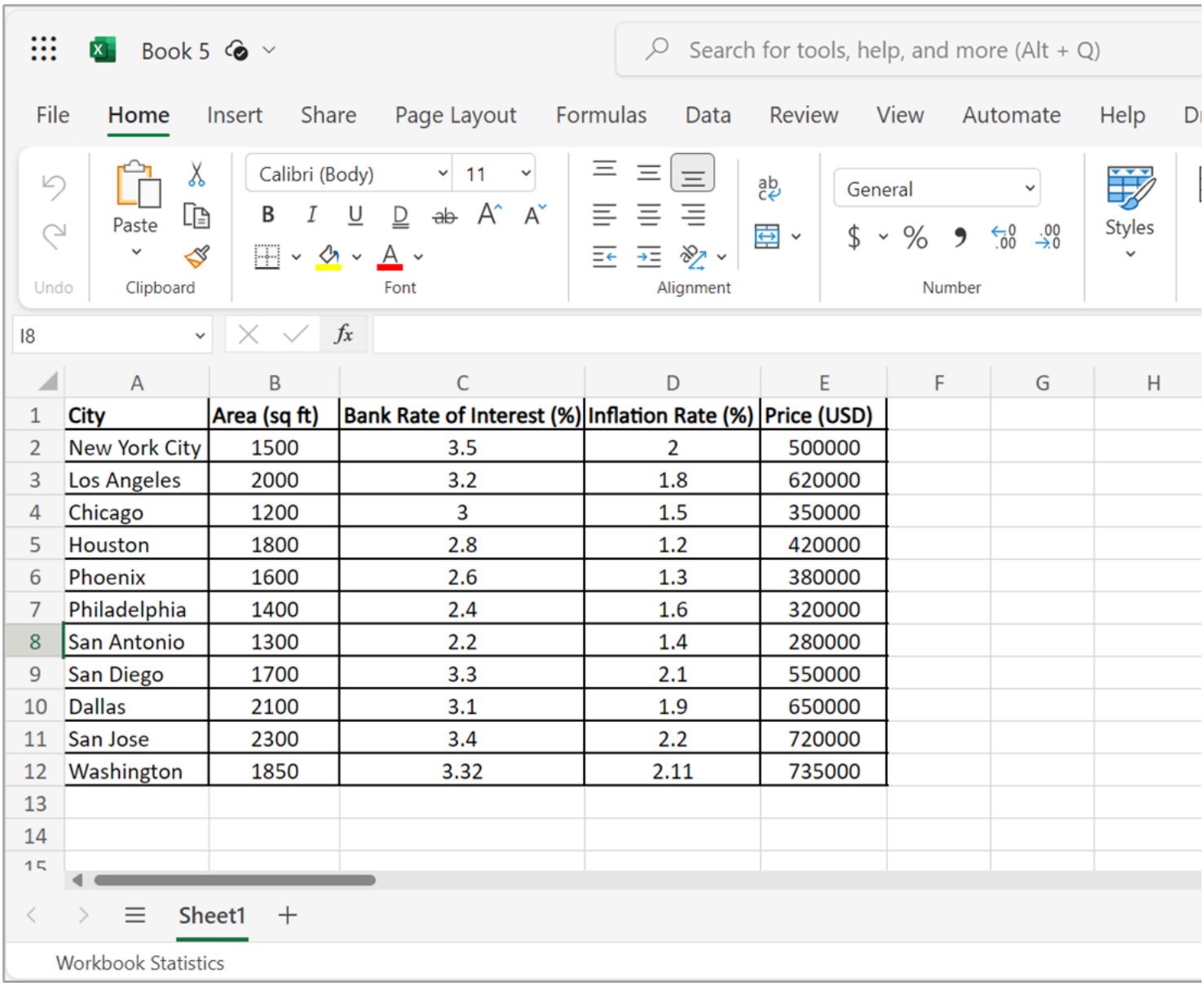
13								
14								

2. The **XLMiner Analysis ToolPak** will appear. Select **Linear Regression**.



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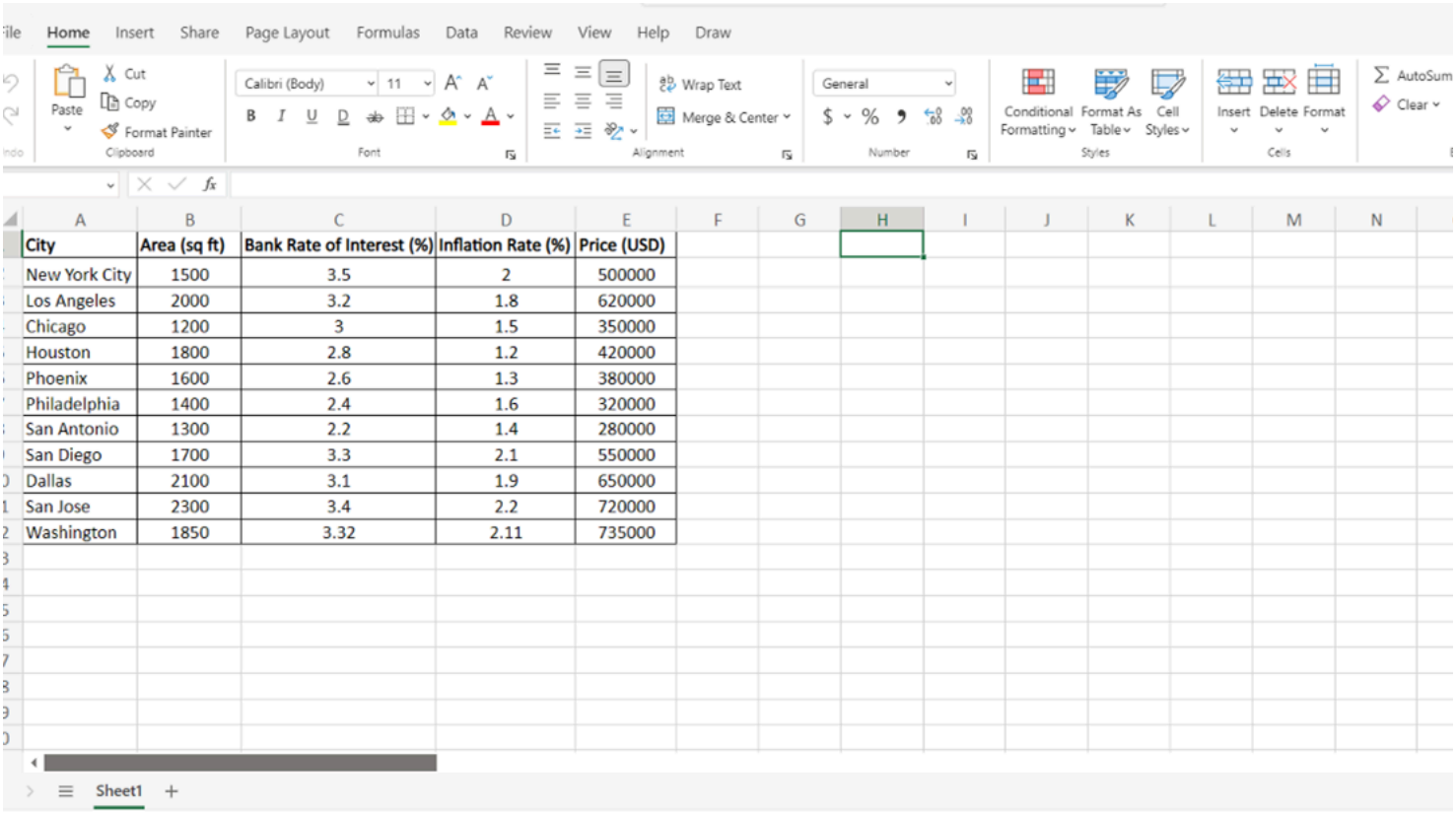
3. Populate the regression analysis fields with the below range.

- **Input Y Range:** The housing prices **($E$1:$E$12)**.
- **Input X Range:** Area in square feet, the inflation rate, and the bank's rate of interest **($B$1:$D$12)**.
- Select the **Labels** checkbox.
- Uncheck **Confidence Level**.
- **Output Range:** $H$1
- Select **OK**.



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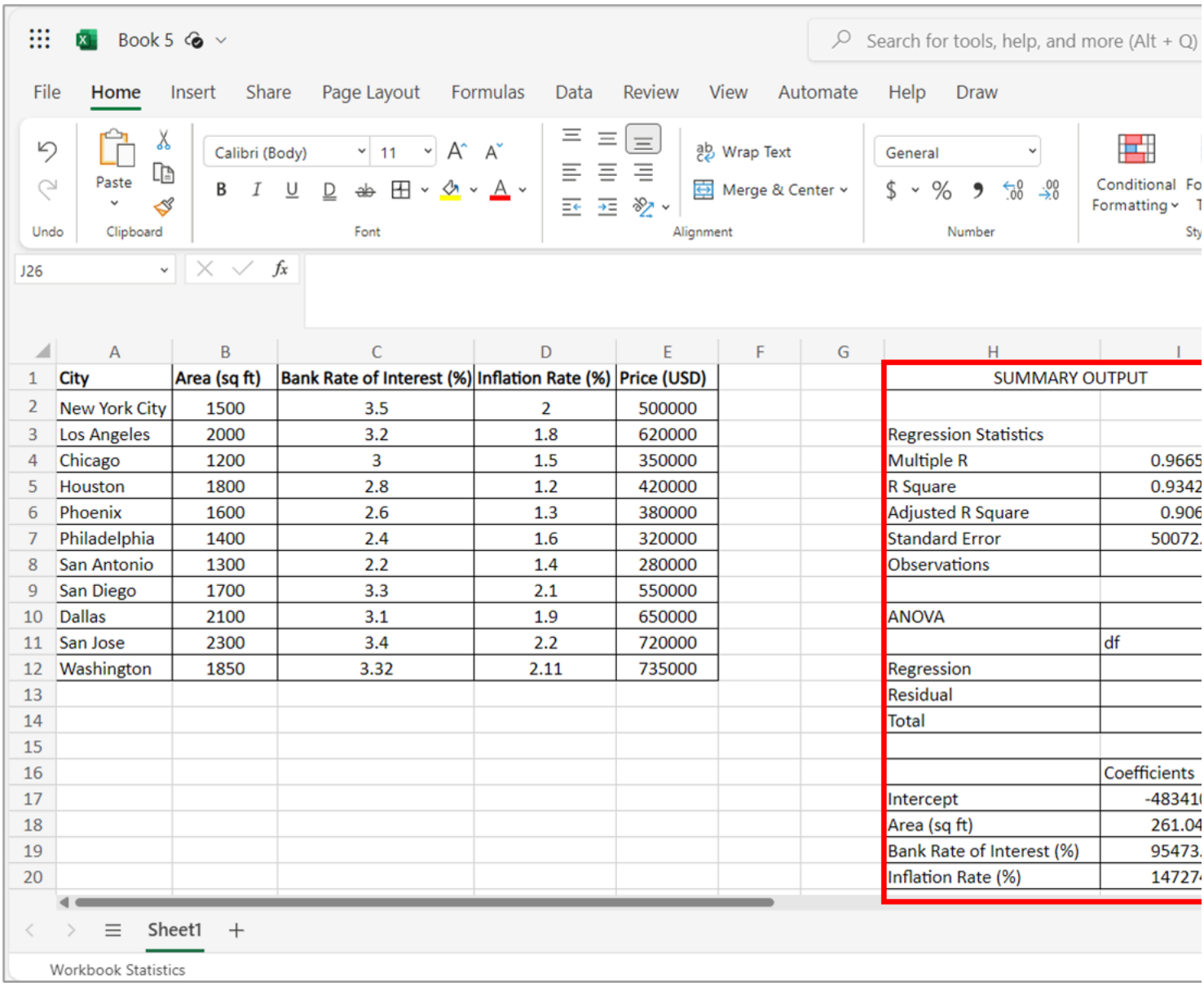
4. The regression analysis results will appear.



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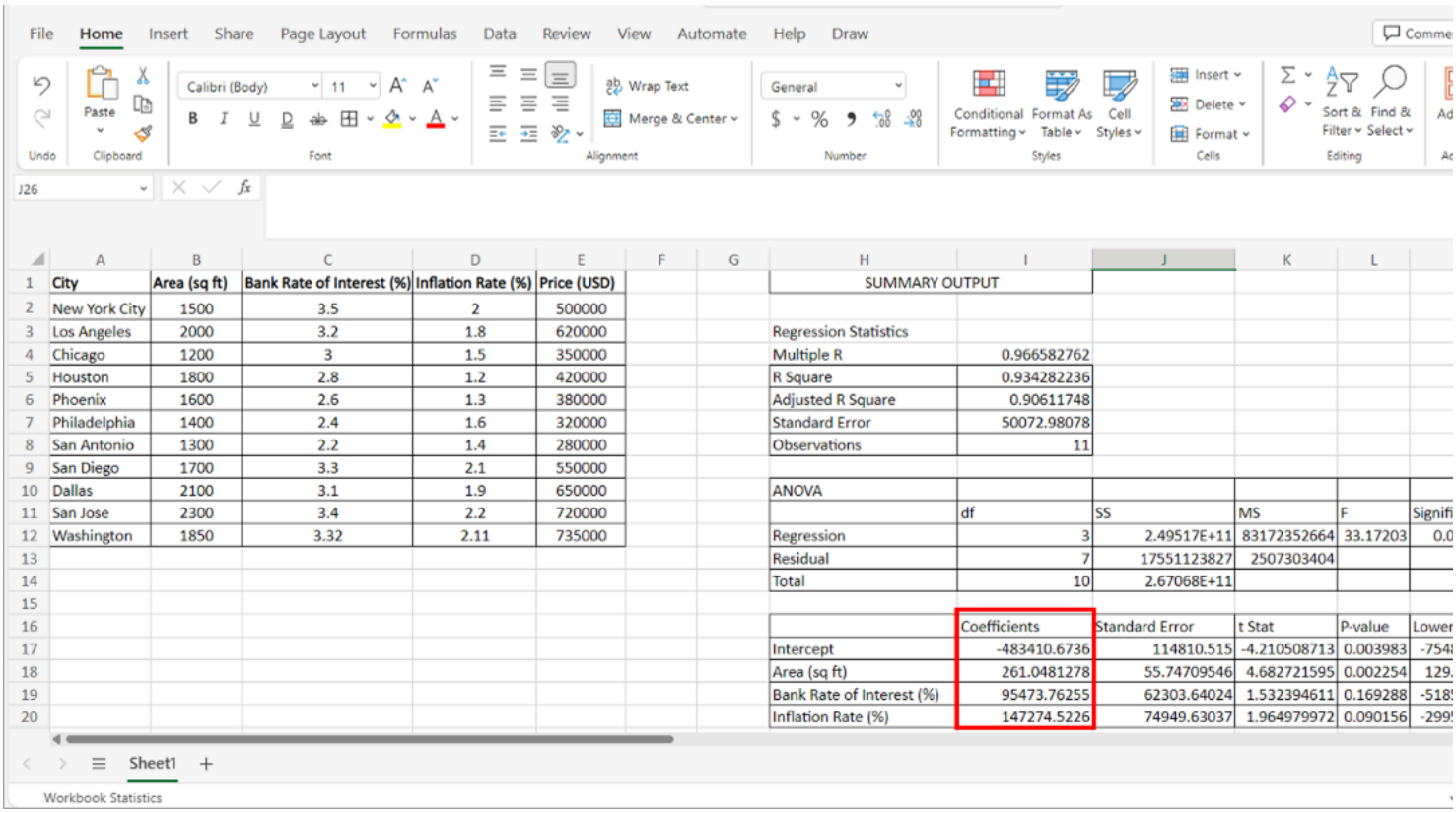
SUMMARY OUTPUT	
Regression Statistics	
Multiple R	0.9665
R Square	0.9342
Adjusted R Square	0.906
Standard Error	50072.
Observations	
ANOVA	
	df
Regression	
Residual	
Total	
	Coefficients
Intercept	-48341
Area (sq ft)	261.04
Bank Rate of Interest (%)	95473.
Inflation Rate (%)	14727

5. The regression output shows the coefficients. You can thus determine the regression equation as follows:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$$
$$Y = -483410.67 + 261.05X_1 + 95473.76X_2 + 147274.52\ X_3$$



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SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.966582762
R Square	0.934282236
Adjusted R Square	0.90611748
Standard Error	50072.98078
Observations	11

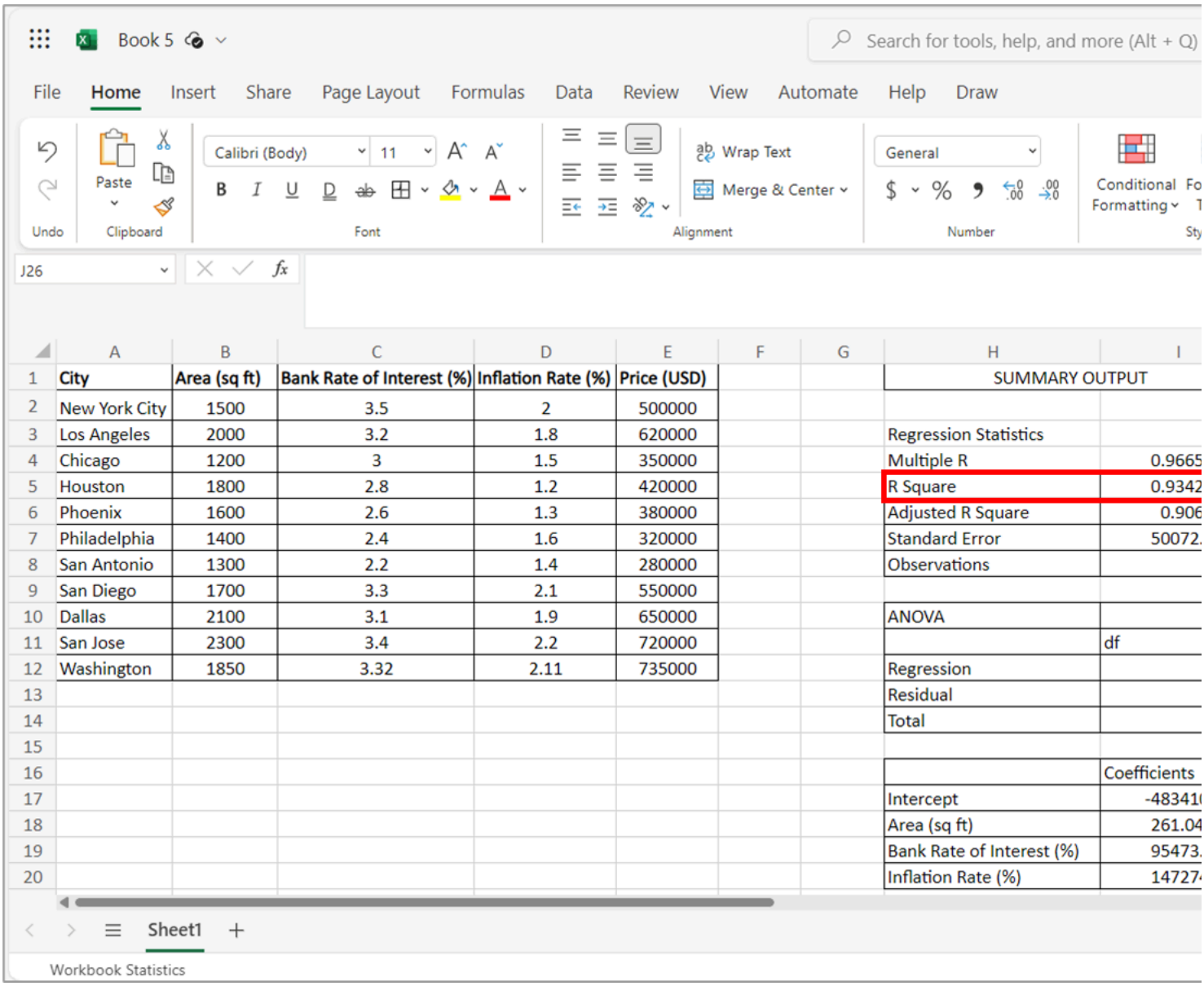
ANOVA	df	SS	MS	F	Signifi
Regression	3	2.49517E+11	83172352664	33.17203	0.0
Residual	7	17551123827	2507303404		
Total	10	2.67068E+11			

	Coefficients	Standard Error	t Stat	P-value	Lower
Intercept	-483410.6736	114810.515	-4.210508713	0.003983	-754
Area (sq ft)	261.0481278	55.74709546	4.682721595	0.002254	129.
Bank Rate of Interest (%)	95473.76255	62303.64024	1.532394611	0.169288	-518
Inflation Rate (%)	147274.5226	74949.63037	1.964979972	0.090156	-299

Task 3: Evaluate the model

Next, you'll evaluate the regression model by examining some of the values in this analysis summary.

1. Examine the **R-squared value**. The R-squared value for the given data set is 0.93. This means that the independent variables, the area in square feet, the inflation rate, and the bank's rate of interest, are collectively responsible for 93% of the variance in the house price.



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	H	I
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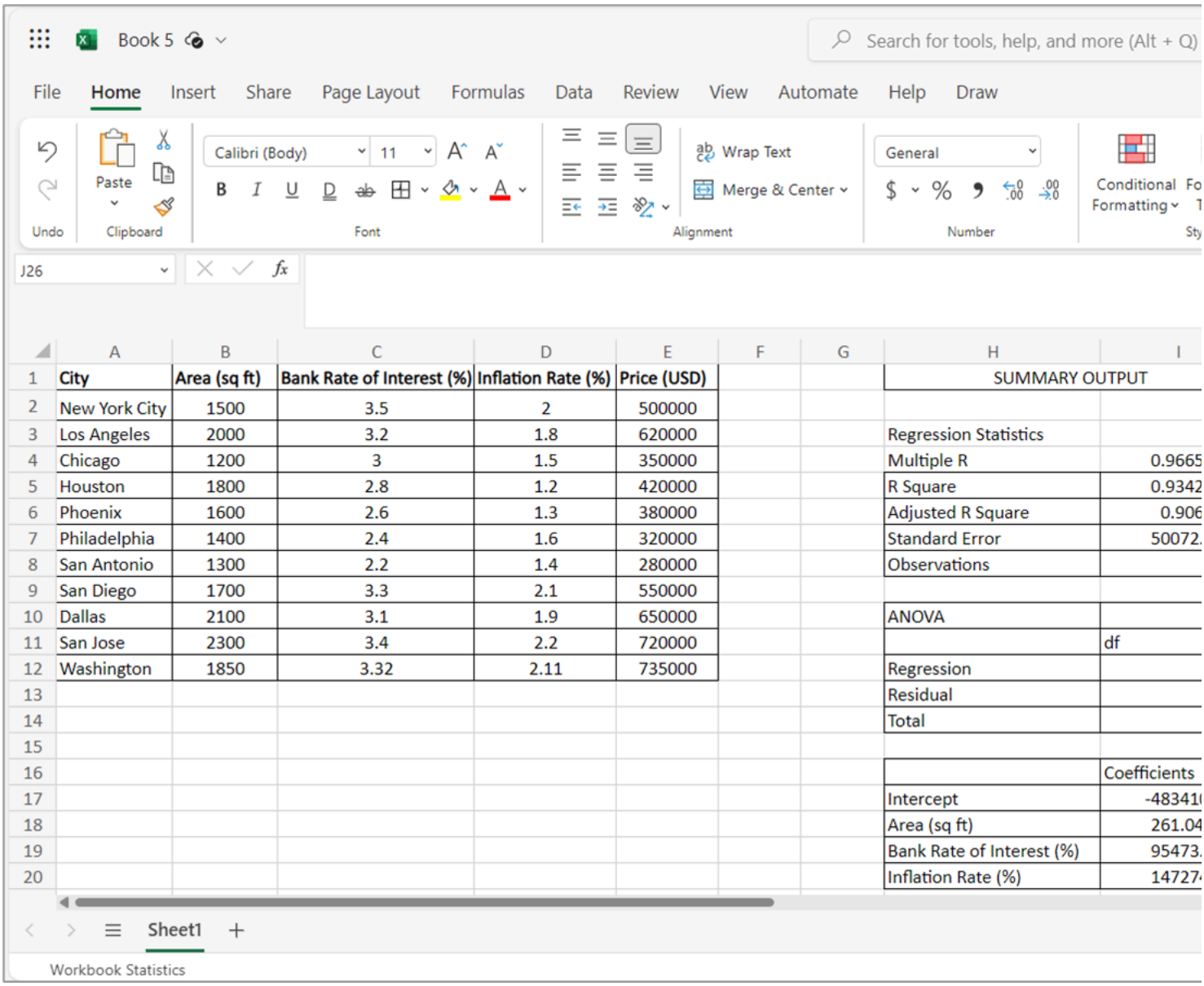
		Coefficients
	Intercept	-48341
	Area (sq ft)	261.04
	Bank Rate of Interest (%)	95473.
	Inflation Rate (%)	14727

2. Next, assess the statistical significance of the regression sum of squares. Examine the ANOVA table produced for the given data set using Excel.

This table shows the **F-statistic** and the **Significance F** value. The F-statistic value is 33.17, which is a high value, and the model's p-value or Significance F is 0.000165, which is less than 0.05. This indicates that one or all of the independent variables have explanatory power beyond what would be expected by chance.



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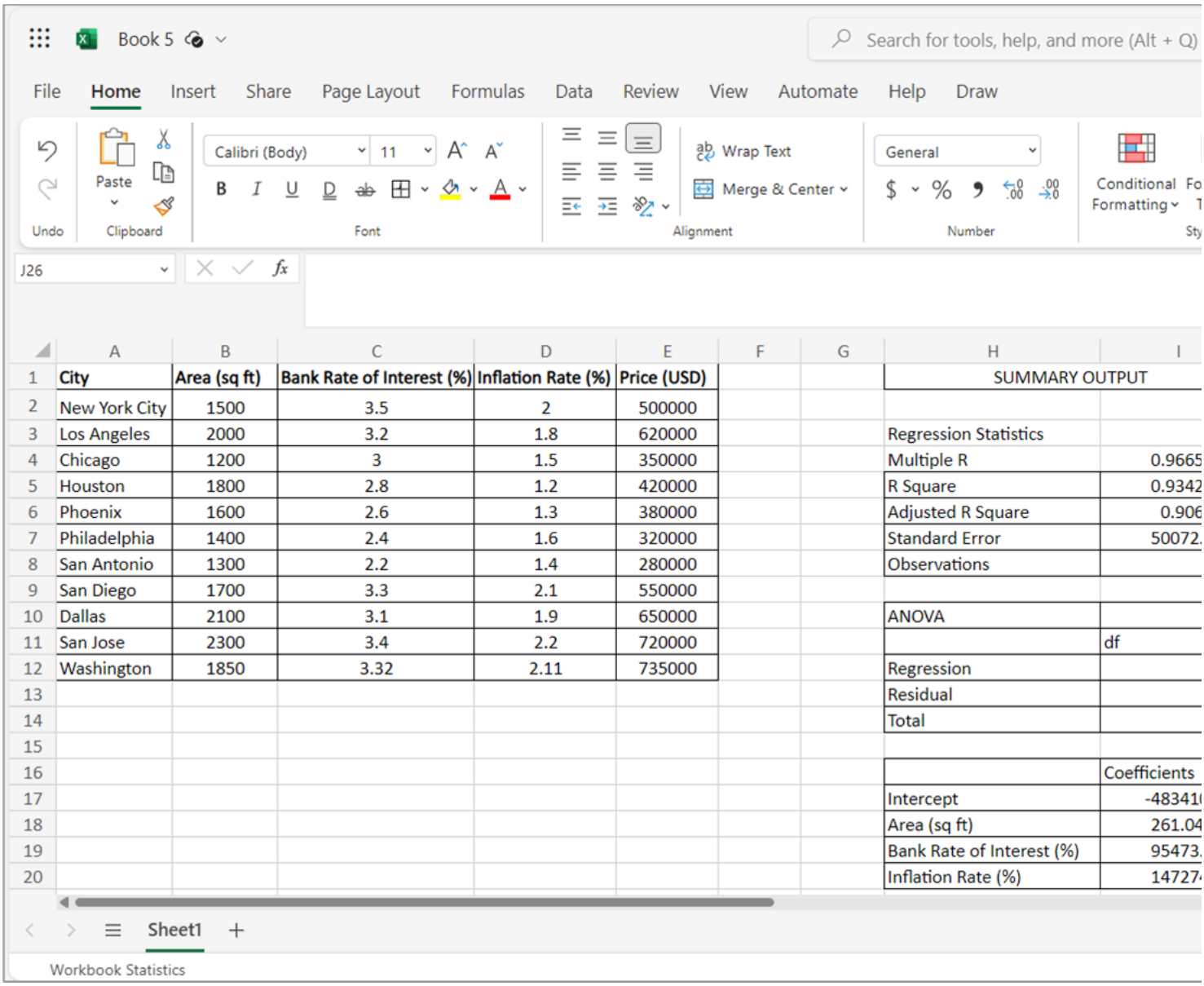
3. To assess the contribution of each independent variable, examine their **p-values**. The p-values for the area in square feet is 0.0022, the bank's rate of interest is 0.16, and that of the inflation rate is 0.09. A low p-value, typically less than 0.05, indicates that the corresponding regression coefficient is statistically significant, implying that the independent variable has a meaningful impact on the dependent variable.



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9	San Diego	1700	3.3	2.1	550000				
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12	Washington	1850	3.32	2.11	735000			Regression	
13								Residual	
14								Total	
15									
16									Coefficients
17								Intercept	-48341
18								Area (sq ft)	261.04
19								Bank Rate of Interest (%)	95473.
20								Inflation Rate (%)	14727

Congratulations! You've completed the lab activity on multiple linear regression, and you are now ready to move to the next topic.

Author(s)

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- Sangeeta Srinivasan
- Rashi Kapoor